

Falcon: Federated Incremental Learning Framework for Intrusion Detection in Heterogeneous Consumer-Centric Internet of Things

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Abstract—The growing heterogeneity of consumer-centric IoT and the continuous evolution of cyber threats are increasingly rendering traditional intrusion detection systems insufficient. Such systems suffer from data silos, severe class imbalance, and limited adaptability to novel attacks. To address these challenges, we propose Falcon, a federated incremental learning framework for intrusion detection in heterogeneous consumer-centric IoT traffic. Falcon enables collaborative training across multiple clients and remains effective under *non-IID* client data distributions and concept drift. It combines a hybrid data-balancing scheme with a CNN-BiLSTM-Attention backbone for learning rich, class-balanced representations, and a drift-triggered incremental module that integrates KL-based drift detection with a dual-head classifier, memory replay, and multi-teacher distillation to learn emerging attacks without catastrophic forgetting. Experiments on three benchmark datasets show that Falcon achieves absolute Precision gains of up to 20.76% over strong centralized and federated baselines in both binary and multi-class tasks.

Index Terms—Federated learning, Incremental learning, Intrusion detection, Consumer-centric internet of things, Imbalanced classification, Knowledge distillation, Memory replay, Distribution drift.

I. INTRODUCTION

SMART wearables and consumer electronics (e.g., smartwatches, AR/VR headsets, home gateways, and healthcare devices) are increasingly pervasive and continuously interact with smartphones, residential networks, and cloud services. Such connectivity enlarges the attack surface and makes intrusion detection essential. Deep learning-based IDS has therefore become an important solution due to its ability to learn high-dimensional traffic representations [1], [2], [3]. However, practical deployment is constrained by privacy requirements that prevent raw traffic from being centrally aggregated, resulting in fragmented data silos and limited local coverage [4].

However, practical IDS deployment for consumer devices is constrained by strict privacy and compliance requirements (e.g., health-related data and fine-grained user-behavior traces), which typically prohibit raw data from being centrally

aggregated across vendors, service providers, or user domains. This results in fragmented data silos and limited coverage at each participant, making it difficult to build globally consistent detectors under privacy-preserving constraints [4].

Federated learning (FL) provides a promising paradigm for privacy-preserving collaboration [5]. Participants (e.g., device cohorts, smartphone proxies, and edge sites) keep raw data locally and only exchange model updates during collaborative training. In this manner, statistical knowledge from heterogeneous multi-source device traffic can be aggregated without directly violating data-sharing restrictions, alleviating siloed learning and improving generalization of the resulting global model.

Nevertheless, deploying IDS within an FL pipeline does not automatically resolve the core learning challenges in consumer-centric IoT ecosystems. First, local data are typically severely imbalanced: benign synchronization and routine service traffic dominate, while intrusions are rare, diverse, and often long-tailed [6]. Such skewed local distributions can bias federated aggregation toward the majority benign behavior, causing poor sensitivity to minority intrusions [7]. Second, consumer-centric IoT traffic is high-dimensional, temporally dependent, and structurally complex. A single deep architecture often struggles to simultaneously capture short-term burst patterns (e.g., repeated pairing attempts), long-range temporal dependencies (e.g., slow-moving compromise), and fine-grained anomaly cues. Third, real-world CIoT is inherently dynamic: shifting user behavior, device mobility, and the continuous emergence of novel attack variants induce distribution drift, degrading the effectiveness of static global models even under privacy-preserving training.

These observations indicate that simply porting a centralized IDS into an FL setting is insufficient. A practical FL-based IDS for CIoT should satisfy three requirements: (i) *re-balancing* highly skewed local datasets to avoid collapsing into benign-dominated decision boundaries, (ii) learning *rich and stable representations* that capture both local patterns and long-range temporal dependencies under heterogeneous clients, and (iii) enabling *incremental adaptation* to distribution drift and novel

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intrusions without catastrophic forgetting.

Motivated by these requirements, we propose **Falcon**—a Federated incremental framework for intrusion detection in heterogeneous consumer-centric IoT. Falcon advances along three lines: *data rebalancing*, *representation modeling*, and *incremental adaptation*. In the federated stage, Falcon jointly employs data-layer rebalancing and Focal Loss to increase both the effective sampling frequency and gradient contribution of minority intrusion classes, mitigating benign-dominant bias. A hybrid *CNN-BiLSTM-Attention* backbone is constructed to capture local patterns, temporal dependencies, and salient segments in traffic sequences. To handle evolving environments, a sliding window with KL divergence detects distribution drift and triggers incremental updates. During incremental adaptation, Falcon integrates a dual-head classifier, memory replay, and multi-teacher knowledge distillation to balance stability and plasticity, absorbing new intrusion behaviors while preserving performance on previously learned classes. Compared with existing federated intrusion detection and incremental learning approaches, Falcon introduces several key novelties. First, it provides a unified framework that jointly addresses class imbalance, representation learning, and incremental adaptation under heterogeneous IoT environments. Second, it incorporates a drift-triggered incremental learning mechanism, enabling the model to dynamically adapt to evolving attack patterns. Third, it integrates multi-teacher knowledge distillation with memory replay, effectively balancing stability and plasticity during incremental updates. The main contributions of this work are summarized as follows:

- We propose a federated rebalancing strategy that combines balanced sampling and Focal Loss, jointly reshaping the effective sample distribution and the gradient contributions of minority classes. This strategy improves F1-Score by up to **97.96%** and Recall by up to **89.32%** over vanilla federated learning on imbalanced intrusion datasets.
- We design a CNN-BiLSTM-Attention feature extractor to capture local patterns, bidirectional temporal dependencies, and salient segments, enabling more discriminative representations under heterogeneous clients. Compared to a vanilla LSTM baseline, this architecture achieves an **88.37%** improvement in Accuracy and a **95.80%** improvement in F1-Score on severely imbalanced multi-class tasks.
- We develop a drift-triggered incremental adaptation mechanism that integrates sliding-window KL-based drift detection with a dual-head classifier, memory replay, and multi-teacher distillation. This mechanism enables cross-dataset incremental learning with up to **90%** relative improvement in adaptation performance while maintaining over **96%** knowledge retention on previously learned classes.

The remainder of this paper is organized as follows. Section II reviews related work on intrusion detection for consumer devices, federated IDS, imbalanced learning, and incremental learning. Section III presents the proposed Falcon framework, including rebalancing with Focal Loss, the CNN-

BiLSTM-Attention model, KL-based drift detection, and federated incremental learning with multi-teacher distillation and memory replay. Section IV reports experimental results on three benchmark datasets, including comparisons with state-of-the-art baselines, cross-dataset incremental learning, and ablation studies. Section V concludes the paper and discusses future directions.

II. RELATED WORK

This section reviews representative studies related to intrusion detection in CIoT, with an emphasis on three threads: (i) deep-learning and federated-learning-based intrusion detection, (ii) imbalanced and long-tailed learning for intrusion detection, and (iii) incremental learning under evolving threats. We highlight the limitations of existing works in heterogeneous and dynamic deployments and position our Falcon framework accordingly.

A. Deep Learning and Federated Learning for Intrusion Detection

Centralized intrusion detection has evolved from signature/statistical approaches to machine learning and deep representation learning. Early signature-based methods can efficiently identify known patterns and remain useful in stable environments. For example, Roughan *et al.* proposed a statistical-signature-based scheme for IP traffic classification [8]. However, such approaches rely on relatively fixed signatures or features and tend to degrade under encrypted traffic, dynamic ports, or rapidly evolving behaviors.

To reduce reliance on plaintext payloads, traditional machine learning methods extract flow-level statistics and observable side-channel features and train classical classifiers. Recent work shows that even under TLS 1.3, encrypted traffic in malware detection settings can be passively characterized using metadata-derived representations (e.g., TLS record-length and direction patterns) together with protocol-aware preprocessing [9]. This enables feature-based classifiers without decrypting payloads. Although such methods improve practicality under encryption, they typically require non-trivial feature engineering and often generalize poorly across domains due to dataset bias and distribution shift.

Deep learning further enables end-to-end representation learning and has become a dominant paradigm in NIDS/IDS. Recent advances also leverage pretraining to enhance representation robustness. For instance, Trafficformer adopts a pre-trained model for traffic data modeling [10]. In consumer-electronics scenarios, deep IDS models have been explored for healthcare/IoMT electronics and medical traffic monitoring [11], [12], as well as vehicular electronics (e.g., VANET-oriented detection) [13]. Despite their strong detection performance in controlled settings, centralized deep IDS generally struggles in real deployments due to privacy constraints and the difficulty of collecting sufficiently labeled and representative data, where labeling itself is a persistent bottleneck [4].

Federated learning (FL) provides a privacy-preserving collaborative training paradigm where network traffic data remain local and only model updates are exchanged. This

paradigm is particularly relevant for CIoT, where sensitive user behaviors and privacy-sensitive traffic traces cannot be easily centralized. Hendaoui *et al.* proposed FLADEN, a federated learning framework for anomaly detection in IoT networks [14]. Olanrewaju-George and Pranggono developed a federated intrusion detection system for IoT using both unsupervised and supervised deep learning models [15], while Beuran *et al.* introduced FedMSE, a semi-supervised federated learning approach for IoT network intrusion detection to further reduce dependence on labeled data [16]. To improve robustness under *non-IID* data, Yan *et al.* studied personalized FL for IoT intrusion detection [17], and Ding *et al.* explored meta-learning-based federated detection for large-scale IoT settings [18]. Recent works in consumer-electronics contexts further investigated federated deep IDS for consumer-centric IoT [19], privacy-preserving designs (e.g., quantum FL) [20], and SDN-assisted federated hybrid IDS frameworks [21].

However, existing FL-based IDS studies still face three fundamental gaps in consumer-centric IoT. First, client data are often severely imbalanced and long-tailed, amplifying the bias of global aggregation. Second, heterogeneous device behaviors and dynamic participation intensify representation instability and distribution drift. Third, many FL-IDS solutions are designed as *static* training pipelines and lack systematic mechanisms for drift-triggered incremental adaptation. Moreover, while long-tailed heterogeneous learning has been studied in consumer electronics (e.g., FedHLC) [22], such advances are not yet fully integrated into an end-to-end federated IDS pipeline that simultaneously handles imbalance, rich temporal representation, and drift-driven incremental updates.

B. Learning with Imbalanced and Long-Tailed Intrusion Data

Class imbalance is ubiquitous in intrusion detection: benign traffic dominates while attacks are rare, diverse, and often long-tailed [6]. Imbalance learning approaches are typically categorized into data-level, algorithm-level, and ensemble-level methods.

Data-level methods rebalance the training distribution via over-/under-sampling. Recent synthetic over-sampling methods generate minority-class samples for imbalanced classification by jointly considering minority and majority class distributions [23], and SMOTE-style resampling has been further tailored to intrusion detection in IoT and sensor networks. Recent work on WSN intrusion detection integrates SMOTE with Tomek links (SMOTE-Tomek), which synthesizes minority instances while removing borderline/overlapping samples, yielding a cleaner and more balanced training set for classical machine-learning detectors [24]. Recent studies have further integrated synthetic sample generation with deep representation learning to improve intrusion detection under severe class imbalance [25]. Despite their effectiveness, oversampling-based methods may still introduce redundant synthetic samples and increase the risk of overfitting, while undersampling may discard informative majority-class samples, both of which can be detrimental under complex attack mixtures.

Algorithm-level methods improve the learning process to better emphasize minority classes. Recent optimization-aware

intrusion detection approaches have combined imbalance-aware sampling, feature embedding, and classifier design to mitigate the adverse effect of skewed data distributions [26]. However, such methods may still be sensitive to optimization settings and training stability under highly heterogeneous attack scenarios. More generally, class-imbalance learning in FL has been recognized as a key challenge, as skewed client distributions can bias global updates [7]. Recent FL methods explicitly target imbalanced intrusion detection, such as undersampling-aware learning and aggregation strategies [27], yet they often do not jointly address representation modeling and incremental adaptation under drift.

Ensemble methods combine multiple learners to improve robustness under imbalance. Evolutionary and genetic-programming-based classifiers have also been explored for imbalanced data classification, including recent distance-based genetic programming approaches designed specifically for imbalanced binary settings [28]. For example, boosting-based IDS systems [29] and recent SMOTE-style imbalance-aware IDS designs [30] can enhance stability but may incur higher training cost and be less suitable for dynamic deployment environments.

Overall, although imbalance mitigation has been widely studied, existing solutions are often applied as isolated components. In consumer-centric IoT, imbalance learning must be coupled with strong temporal representation modeling and drift-aware adaptation to maintain sustained effectiveness.

C. Incremental Intrusion Detection under Evolving Threats

Real-world CIoT evolves rapidly due to shifting user behavior, device mobility, and changing device participation. These factors induce concept drift and are often accompanied by the emergence of novel attacks. A common industrial practice is periodic full retraining using historical repositories [31], but this is computationally expensive and often impractical due to high computation/communication costs, privacy constraints, and limited storage. Moreover, the continuous emergence of novel attacks has motivated intelligent cybersecurity frameworks for adaptive defense [32].

Incremental learning aims to update models with newly arriving data (e.g., emerging attack types or shifted traffic distributions) while preserving previously learned knowledge, which is essential for intrusion detection under concept drift. Sun *et al.* proposed temporal-incremental learning for Android malware detection to handle evolving families and drift [33], yet catastrophic forgetting remains a central challenge without principled stability-plasticity control. In traffic modeling, representation advances (e.g., transformer-style multi-level representations) highlight the potential of richer sequence modeling under evolving behaviors [34]. In federated settings, incremental IDS pipelines have also been explored [35], but many studies either lack robust drift triggering or do not explicitly address long-tail imbalance together with incremental updates. Recent generative few-shot IDS methods further attempt to mitigate data scarcity during adaptation and improve trustworthiness via explainability [36], but integrating such ideas into an FL pipeline still requires careful design to avoid instability and privacy leakage.

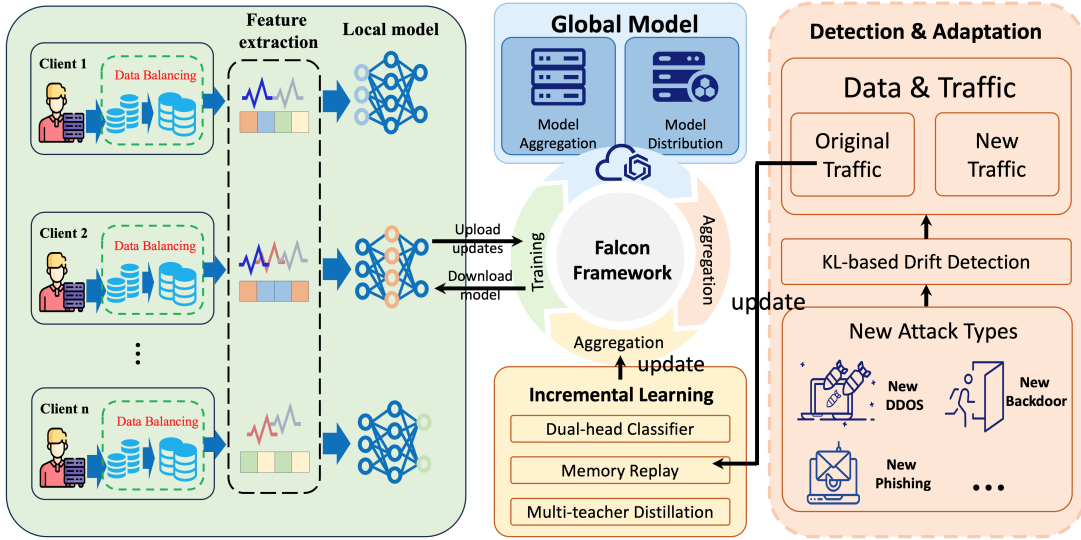


Fig. 1. Overall architecture of the proposed Falcon framework. Clients perform data balancing and feature extraction, train local models, and collaboratively contribute to the global model through federated aggregation, while a drift detector monitors incoming traffic and triggers incremental learning to adapt to emerging attacks.

III. METHOD

In this section, we present the overall design of Falcon, which consists of four stages: (1) local data balancing on each client's network traffic; (2) local training of the CNN-BiLSTM-Attention model on the balanced data; (3) sliding-window drift detection for emerging attack patterns; and (4) incremental global model updating at the central server. Among these, stages (1)–(2) correspond to the federated local training process, while stages (3)–(4) represent the incremental adaptation mechanism. The overall workflow is illustrated in Fig. 1.

A. Balanced Sampler and Focal Loss Function

To address severe class imbalance, we combine a **balanced sampler** and **Focal Loss** to jointly mitigate sample and gradient imbalance. The sampler increases minority-class frequency, while Focal Loss emphasizes hard samples. The overall training procedure is shown in **Algorithm 1**, and experimental results demonstrate significant improvements in minority-class detection.

First, at the sampling level, suppose the dataset contains K classes, denoted as $\mathcal{C} = \{1, 2, \dots, K\}$, and each class k has n_k samples. The objective of the oversampling balancer is to ensure that the number of samples per class in each training batch remains balanced. Specifically, given a batch size B , the number of samples drawn from each class k is defined as

$$B_k \approx \frac{B}{K}. \quad (1)$$

If the number of available samples in a particular class is insufficient, sampling with replacement is applied to that class to ensure that the batch composition satisfies:

$$\sum_{k=1}^K B_k = B, \quad \text{and} \quad B_k \geq 1, \quad \forall k. \quad (2)$$

At the optimization level, we adopt the Focal Loss function instead of the standard cross-entropy loss to further enhance the model's learning capability for hard-to-classify minority samples. For a given sample i belonging to class y_i , let the model's predicted probability for that class be $p_{t,i}$. The traditional cross-entropy loss is defined as:

$$\mathcal{L}_{CE} = -\log p_{t,i}. \quad (3)$$

The Focal Loss, on the other hand, introduces a class-balancing factor α_{y_i} and a focusing parameter γ , defined as:

$$\mathcal{L}_{\text{focal}}(x_i, y_i) = -\alpha_{y_i} (1 - p_{t,i})^\gamma \log(p_{t,i}), \quad (4)$$

where $\alpha_{y_i} \in (0, 1)$ is the class weight (usually larger for minority classes), $\gamma \geq 0$ controls the focusing strength, and $p_{t,i}$ is the model's predicted probability for the true class of sample i .

The total batch loss is then computed as:

$$\mathcal{L}_{\text{batch}} = \frac{1}{B} \sum_{i=1}^B \mathcal{L}_{\text{focal}}(x_i, y_i). \quad (5)$$

By jointly applying the balanced sampler and Focal Loss (as detailed in Algorithm 1), each training iteration ensures that the model sufficiently learns from minority-class samples while concentrating optimization on difficult or misclassified instances. This dual strategy not only alleviates the bias caused by class imbalance but also mitigates overfitting risks introduced by oversampling, thereby enhancing the model's generalization ability and detection performance in highly imbalanced network intrusion detection scenarios.

B. Deep Model Integrating CNN-BiLSTM and Attention Mechanism

To effectively identify evolving and fine-grained attack behaviors in network traffic, it is crucial to capture both spatial

Algorithm 1: Balanced Sampler + Focal Loss Training (BS-FL)

Input: Dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, batch size B , epochs E , model f_θ , optimizer $\mathcal{O}$, Focal Loss params (α, γ) , grad clip c ; *sampler option*: either **BalancedBatchSampler** (per-class balanced mini-batches) or **WeightedRandomSampler** (inverse-frequency weights).

Output: Trained parameters θ^* .

```

1 Build label indices: group sample indices by class to
  obtain  $\{\mathcal{I}_k\}_{k=1}^K$ ;
2 if BalancedBatchSampler then
3   Compute per-class quota  $q_k \leftarrow \lfloor B/K \rfloor$ ; distribute the
  remainder  $B \bmod K$  to the first classes;
4   Initialize sampler that yields batches with
   $\sum_k q_k = B$  (with replacement if a class is scarce);
5 else // WeightedRandomSampler
6    $n_k \leftarrow |\mathcal{I}_k|$ ; set per-sample weight  $w_i \propto 1/n_{y_i}$ ;
7   Initialize sampler with weights  $\{w_i\}_{i=1}^N$  to draw  $B$ 
  i.i.d. indices per step;
8
9 for  $e \leftarrow 1$  to  $E$  do
10  foreach batch indices  $\mathcal{B}$  yielded by the sampler do
11    Load mini-batch  $(X_{\mathcal{B}}, y_{\mathcal{B}})$ ;
12     $z \leftarrow f_\theta(X_{\mathcal{B}})$ ; // logits
13    /* Focal Loss */
14     $\ell_{\text{CE}} \leftarrow \text{CrossEntropy}(z, y_{\mathcal{B}})$ ; // per-sample
15     $p_t \leftarrow \exp(-\ell_{\text{CE}})$ ;
16     $\ell_{\text{Focal}} \leftarrow \alpha \cdot (1 - p_t)^\gamma \cdot \ell_{\text{CE}}$ ;
17     $L \leftarrow \text{mean}(\ell_{\text{Focal}})$ ;
18     $\mathcal{O}.\text{zero\_grad}()$ ; backprop  $L$ ;
19     $\mathcal{O}.\text{clip\_grad\_norm}(\theta, c)$ ; // e.g.,  $c = 2.0$ 
20     $\mathcal{O}.\text{step}()$ ;
21 return  $\theta^* \leftarrow \theta$ ;

```

and temporal dependencies embedded in packet sequences. Convolutional neural networks excel at extracting local spatial patterns, such as statistical and structural features within packets, but lack the ability to model long-range temporal relationships. In contrast, bidirectional long short-term memory networks (BiLSTMs) are capable of learning sequential dependencies and contextual transitions over time, yet they often treat all temporal features with equal importance. The attention mechanism further complements these models by adaptively assigning higher weights to critical traffic features, thereby enhancing the interpretability and focusing capability of the model. Therefore, combining CNN, BiLSTM, and attention allows the model to jointly learn hierarchical representations—spatial features through convolution, temporal dependencies through recurrent connections, and feature saliency through attention—resulting in a more comprehensive understanding of dynamic network behaviors, as illustrated in Fig. 2.

Specifically, the model input is a temporal feature sequence

of length T , denoted as

$$X = [x_1, x_2, \dots, x_T], \quad (6)$$

where each time-step input $x_t \in \mathbb{R}^F$, and F represents the feature dimension.

First, in the local feature extraction stage, a one-dimensional convolutional layer is applied to slide along the temporal dimension. Suppose there are D convolutional kernels, with weights

$$W^{(k)} \in \mathbb{R}^{K \times F} \quad (7)$$

and kernel width K . The output of the k -th kernel at position t is given by:

$$c_t^{(k)} = \sigma \left(\sum_{j=0}^{K-1} W_j^{(k)} \cdot x_{t+j} + b^{(k)} \right), \quad (8)$$

where $\sigma(\cdot)$ denotes the activation function and $b^{(k)}$ is the bias term. After concatenating all kernel outputs, the feature map sequence is obtained as

$$C = [c_1, c_2, \dots, c_{T'}], \quad (9)$$

where $c_t \in \mathbb{R}^D$ and $T' = T - K + 1$.

Next, the feature map sequence is fed into a bidirectional LSTM (BiLSTM) layer to model long-range temporal dependencies. For each time step t , the forward LSTM recursively computes the hidden state:

$$\vec{h}_t, \vec{c}_t = \text{LSTM}^f(x_t, h_{t-1}, c_{t-1}), \quad (10)$$

and the backward LSTM recursively computes the backward hidden state:

$$\overleftarrow{h}_t, \overleftarrow{c}_t = \text{LSTM}^b(x_t, h_{t+1}, c_{t+1}). \quad (11)$$

The forward and backward hidden states are concatenated to obtain the comprehensive output at time t :

$$h_t = [\vec{h}_t^{(f)}; \overleftarrow{h}_t^{(b)}], \quad (12)$$

where $h_t \in \mathbb{R}^{2H}$ and H is the number of hidden units in a single-direction LSTM.

On this basis, a self-attention mechanism is introduced to adaptively focus on key time steps. Specifically, each BiLSTM output h_t is linearly transformed into query, key, and value vectors:

$$q_t = W_q h_t, \quad k_t = W_k h_t, \quad v_t = W_v h_t, \quad (13)$$

where W_q , W_k , and W_v are trainable parameter matrices.

The similarity between each query q_t and all keys k_j is computed to obtain unnormalized attention scores:

$$s_{tj} = \frac{q_t^\top k_j}{\sqrt{d_k}}, \quad (14)$$

where d_k is the dimension of the key vector for scaling. The attention weights are obtained by applying the softmax function:

$$\alpha_{tj} = \frac{\exp(s_{tj})}{\sum_{l=1}^{T'} \exp(s_{tl})}. \quad (15)$$

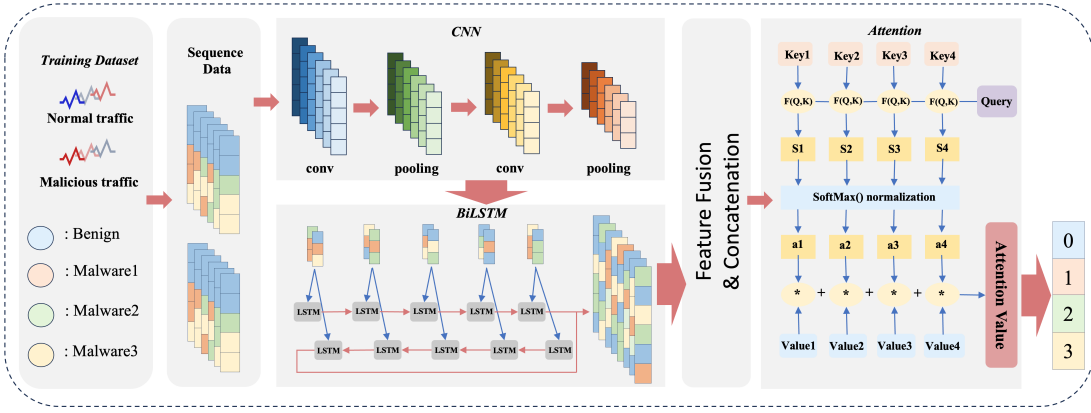


Fig. 2. Pipeline of the proposed CNN-BiLSTM-Attention model for network traffic classification. Raw traffic is converted into multivariate temporal sequences and fed to a 1D-CNN to extract local patterns, followed by a BiLSTM to capture bidirectional temporal dependencies and a self-attention module to aggregate salient features before final classification.

Each query's aggregated representation z_t is obtained as a weighted sum of the value vectors:

$$z_t = \sum_{j=1}^{T'} \alpha_{tj} v_j. \quad (16)$$

To obtain a global representation vector of the sequence, all z_t are pooled:

$$z = \frac{1}{T'} \sum_{t=1}^{T'} z_t. \quad (17)$$

Finally, the aggregated global feature vector z is fed into a fully connected layer to output multi-class prediction probabilities:

$$\hat{y} = \text{softmax}(W_o z + b_o), \quad (18)$$

where W_o and b_o are the output layer parameters.

Through this architecture, the model first extracts local short-term features via CNN, captures global temporal dependencies via BiLSTM, and then adaptively integrates information across different time steps via the self-attention mechanism. This significantly enhances the model's capability to recognize and classify critical behavioral segments in complex and imbalanced network traffic scenarios.

C. Data Drift Detection

To enable the recognition of novel attacks and support incremental learning of network traffic, we introduce a sliding window-based data drift detection mechanism, as described in **Algorithm 2**. This mechanism continuously monitors whether data drift occurs within the current window of network traffic, providing a foundation for subsequent federated incremental learning.

Specifically, the system maintains a sliding window containing network traffic samples over several time slices. For this window, we extract the packet length sequences and direction sequences of all flows, concatenate these features into one-dimensional vectors, and then compute the normalized probability distributions for each feature type.

Algorithm 2: Sliding-Window Data Drift Detection Based on KL Divergence

Input: Traffic stream $\mathcal{S}$ ordered by time, window size W , step size s , threshold τ , bin size B .
Output: Drift decisions $\{\text{Drift}_t\}$ for each window.

- 1 **Initialize reference window:**
- 2 Extract the first W samples from $\mathcal{S}$ to form $\mathcal{D}_{\text{ref}}$;
- 3 Extract packet-length and direction sequences ($\mathbf{L}_{\text{ref}}, \mathbf{D}_{\text{ref}}$) from $\mathcal{D}_{\text{ref}}$;
- 4 Build histograms $\mathbf{h}_{\text{ref}}^L, \mathbf{h}_{\text{ref}}^D$ with B bins and apply Laplace smoothing to obtain $P_{\text{ref}}^L, P_{\text{ref}}^D$;
- 5 **Divergence computation:**
- 6 $\text{KL}_{\text{len}} \leftarrow \sum_i P_{\text{ref},i}^L \log \frac{P_{\text{ref},i}^L}{Q_{\text{cur},i}^L}$;
- 7 $\text{KL}_{\text{dir}} \leftarrow \sum_i P_{\text{ref},i}^D \log \frac{P_{\text{ref},i}^D}{Q_{\text{cur},i}^D}$;
- 8 $\text{KL}_{\text{tot}} \leftarrow \frac{1}{2} (\text{KL}_{\text{len}} + \text{KL}_{\text{dir}})$;
- 9 **(Optional) Robustness check:**
- 10 $M^L \leftarrow \frac{1}{2} (P_{\text{ref}}^L + Q_{\text{cur}}^L), \quad M^D \leftarrow \frac{1}{2} (P_{\text{ref}}^D + Q_{\text{cur}}^D)$;
- 11 $\text{JS} \leftarrow \frac{1}{2} [\text{KL}(P_{\text{ref}}^L \| M^L) + \text{KL}(Q_{\text{cur}}^L \| M^L)] + \frac{1}{2} [\text{KL}(P_{\text{ref}}^D \| M^D) + \text{KL}(Q_{\text{cur}}^D \| M^D)]$; // backup divergence
- 12 **if** $\text{KL}_{\text{tot}} > \tau$ **then**
- 13 $\text{Drift}_t \leftarrow \text{True}$;
- 14 $\mathcal{D}_{\text{ref}} \leftarrow \mathcal{D}_{\text{cur}}$; // update reference window
- 15 $P_{\text{ref}}^L, P_{\text{ref}}^D \leftarrow Q_{\text{cur}}^L, Q_{\text{cur}}^D$;
- 16 **else**
- 17 $\text{Drift}_t \leftarrow \text{False}$;

To quantify changes in the data distribution, we employ the Kullback–Leibler (KL) divergence:

$$D_{\text{KL}}(P \| Q) = \sum_{i=1}^B P(i) \log \frac{P(i)}{Q(i)}, \quad (19)$$

where P and Q represent the normalized probability distributions of the reference window and the current window,

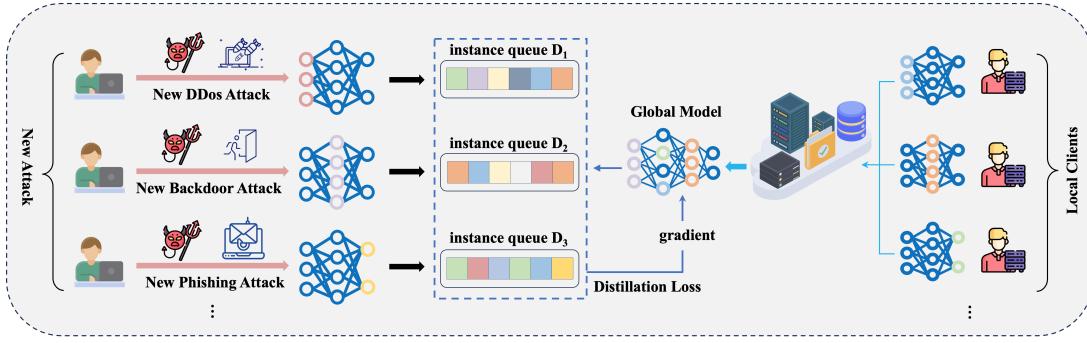


Fig. 3. Federated incremental learning with knowledge distillation for new attack adaptation. When novel attacks are detected, each client locally trains on new traffic and maintains an instance queue D_i of representative samples, while the global server aggregates client updates and applies distillation to adapt the global model without catastrophic forgetting.

respectively. To improve numerical stability, all distributions are smoothed using Laplace smoothing:

$$P'_i = \frac{P_i + \epsilon}{1 + \epsilon B}, \quad Q'_i = \frac{Q_i + \epsilon}{1 + \epsilon B}. \quad (20)$$

The KL divergence is computed separately for the packet length sequences and direction sequences, and their mean is taken as the overall drift measure:

$$D_{KL}^{\text{total}} = \frac{1}{2} (D_{KL}(P'_{\text{len}} \parallel Q'_{\text{len}}) + D_{KL}(P'_{\text{dir}} \parallel Q'_{\text{dir}})). \quad (21)$$

The system periodically slides the window forward and recomputes the feature distributions. When the difference between the current window's feature distribution and the reference distribution exceeds a predefined threshold, a significant data drift is detected, which triggers the incremental learning process.

D. Federated Incremental Learning

When data drift occurs, the federated incremental learning mechanism is triggered, as illustrated in Fig. 3. New samples of traffic classes are collected, and a unified label mapping is applied. In the model's output layer, a dual-classification head structure is adopted: the original classifier handles historical classes, while an additional classifier handles new classes. The outputs of both heads are concatenated for unified multi-class prediction.

To effectively maintain previously acquired knowledge while continuously learning new attack types, the model must overcome catastrophic forgetting, a common issue in incremental learning scenarios. Conventional single-teacher knowledge distillation mitigates this problem by transferring soft-label information from the previous model to the current one, but it struggles to balance old and new knowledge when data distributions shift significantly. To address this limitation, we design a **multi-teacher knowledge distillation** mechanism that integrates both the **historical model** and a **new-class teacher model**. The new-class teacher is trained independently on traffic samples identified by the **data drift detection module (Algorithm 2)**, which continuously monitors recent network flows and detects emerging attack types that deviate from previously seen distributions. This strategy enables the

student model to learn jointly from historical knowledge and newly emerging class semantics, thereby improving adaptability and stability during incremental updates. Furthermore, to reinforce the retention of previously learned representations, a **memory replay** mechanism is incorporated, which periodically reuses a small subset of representative samples from earlier classes during training. For each batch of samples, let the outputs of the historical model, the new-class model, and the student model be denoted by $\mathbf{z}_{\text{old}}$, $\mathbf{z}_{\text{new}}$, and $\mathbf{z}_s$, respectively. The knowledge distillation loss is divided into two parts, corresponding to old-class and new-class data:

$$\mathcal{L}_{\text{old}} = T^2 \text{KL}(\sigma(\mathbf{z}_{\text{old}}/T) \parallel \sigma(\mathbf{z}_s/T)), \quad (22)$$

$$\mathcal{L}_{\text{new}} = T^2 \text{KL}(\sigma(\mathbf{z}_{\text{new}}/T) \parallel \sigma(\mathbf{z}_s/T)), \quad (23)$$

where T is the temperature parameter and $\sigma(\cdot)$ denotes the softmax function.

The overall loss function combines cross-entropy supervision with the two knowledge distillation terms:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{CE}}(\mathbf{z}_s, y) + \beta \mathcal{L}_{\text{old}} + \gamma \mathcal{L}_{\text{new}}, \quad (24)$$

where $\mathcal{L}_{\text{CE}}$ is the standard cross-entropy loss, and α, β, γ are balancing coefficients. This design ensures that the student model acquires discriminative capability for new classes while inheriting knowledge from the historical model and mimicking the new teacher model's performance on new classes.

The memory replay mechanism mixes a portion of original-class samples with new-class samples during training for joint optimization. Let $\mathcal{D}_{\text{new}}$ and $\mathcal{D}_{\text{old}}$ denote the new-class and original-class sample sets, respectively, and let δ be the replay ratio. Then the training set can be expressed as:

$$\mathcal{D} = \mathcal{D}_{\text{new}} \cup \text{replay}(\mathcal{D}_{\text{old}}, \delta). \quad (25)$$

In summary, the proposed federated incremental learning method, combining multi-teacher knowledge distillation and memory replay, leverages dual distillation supervision from historical and new-class models. This approach effectively enhances the model's adaptability to new classes, mitigates catastrophic forgetting, and enables incremental multi-class learning in a federated environment, allowing the model to continuously recognize new attacks.

TABLE I
DATASET INFORMATION

Dataset	Label	Class	Training Samples	IR
USTC-TFC2016	0	Benign	309,887	–
	1	Htbot	6,341	48.87
	2	Nsis-ay	6,165	50.26
	3	Shifu	9,634	32.16
CICIDS2017	0	Benign	923,281	–
	1	DDoS	14,039	65.79
	2	FTP-Patator	3,973	232.45
	3	SSH-Patator	2,979	310.01
CICIDS2018	0	Benign	296,245	–
	1	Botnet Ares	50,946	5.82
	2	Infiltration	106,397	2.78
	3	Web Attack	19	15,591.84

Note: “Label” represents the label assigned to each class of data; “Class” represents the class contained in the dataset; “Training Samples” indicates the number of samples of each class in the training set; “IR” represents the imbalance ratio with respect to the Benign class.

IV. EXPERIMENTS

In this section, we validate the effectiveness of Falcon through empirical experiments. First, we provide detailed experimental settings, including model parameters, evaluation metrics, and an overview of the datasets. Next, we conduct a series of experiments on the proposed Falcon framework, comparing it with several state-of-the-art (SOTA) models and verifying the effectiveness of incremental learning. Finally, ablation studies are designed to demonstrate the effectiveness of each improvement.

A. Experimental Settings

Datasets: Unlike prior works that directly use CSV files, we extract flow features (e.g., packet-length sequences and timestamp sequences) from raw PCAP traces based on 5-tuple aggregation rules. Furthermore, class selection is applied to the processed datasets. The final experimental datasets include the following classes: *USTC-TFC2016* (Benign, Htbot, Nsis-ay, Shifu), *CICIDS2017* (Benign, DDos, FTP-Patator, SSH-Patator), and *CICIDS2018* (Benign, Botnet Ares, Infiltration, Web Attack). Table I summarizes the detailed information of the datasets used in this paper.

Implementation Details: We implemented Falcon using PyTorch and Ray. Focal Loss was adopted with $\alpha = 1.0$ and $\gamma = 2.0$. The number of communication rounds was set to $R = 100$. The server used SGD with momentum, an initial global learning rate of 0.01, and weight decay 1×10^{-4} . Each client performed $E = 5$ local epochs with batch size 128 using AdamW (learning rate 0.01, weight decay 1×10^{-2}), and gradient clipping with a maximum norm of 2.0 was applied for stability.

To simulate heterogeneous federated scenarios, we generated *non-IID* client partitions using a Dirichlet-based allocation strategy, where smaller α_{dir} indicates stronger heterogeneity. We also allowed different clients to have different sample volumes to reflect realistic data imbalance across participants.

Both global and local models used a CNN-BiLSTM-Attention backbone, including 1D convolutional layers with 64

channels and kernel size 3, a two-layer BiLSTM with hidden size 64 and dropout 0.2, an attention module with projection dimension 64, and a classification head ($128 \rightarrow 32 \rightarrow \text{number of classes}$) with ReLU and dropout 0.3.

Baseline and Comparison: For both binary and multi-class classification tasks, the proposed Falcon framework was compared with five state-of-the-art baseline methods, including Fed-ANIDS, FELIDS, FL-IIDS, FedUGI, and FeCoGraph [37], [38], [35], [27], [39]. In addition, we compared the performance of Falcon with TMG-GAN, which was trained locally without a federated learning framework [40].

Evaluation Metrics: Since the datasets are highly imbalanced, accuracy alone is insufficient. We therefore report Accuracy, Precision, Recall, and F1-Score. Precision reflects false-positive control, Recall measures attack detection capability, and F1-Score balances both, making it particularly suitable for imbalanced intrusion detection.

B. Experimental Results

In this section, we present the empirical results of extensive validation experiments conducted on the *USTC-TFC2016*, *CICIDS2017*, and *CICIDS2018* datasets, covering binary classification, multi-class classification, robustness evaluation under client dropout and adversarial attacks, hyperparameter sensitivity analysis, *non-IID* heterogeneity analysis, incremental learning, and ablation studies.

Binary Classification Experiments: In the binary setting, we compare three families of methods: the centralized non-FL detector TMG-GAN, five representative FL-based NIDS (Fed-ANIDS, FELIDS, FL-IIDS, FedUGI and FeCoGraph), and the proposed Falcon framework (denoted as “Ours” in the figures). As shown in Fig. 4, Falcon consistently achieves the best performance on all three datasets and on all four metrics (Accuracy, Precision, Recall and F1-Score), with its curves remaining at the very top of each plot. The superior performance of Falcon can be attributed to its class-rebalancing mechanism. As defined in Eq. (4), the Focal Loss introduces the modulation term $(1 - p_t)^\gamma$, which suppresses the contribution of easy majority-class samples and amplifies the gradients of hard and minority-class samples. Combined with the balanced sampler in Eqs. (1)–(2), Falcon effectively reshapes both the sample distribution and the optimization dynamics. As a result, the model maintains consistently high Precision, Recall, and F1-Score on imbalanced intrusion traffic.

On *USTC-TFC2016*, TMG-GAN and FedUGI obtain relatively high recalls of 95.03% and 94.89%, respectively, but their precisions (91.76% and 95.88%) and F1-Scores (93.36% and 95.36%) are clearly lower than those of Falcon. Falcon reaches 99.72% Accuracy, 99.70% Precision, 99.70% Recall and 99.70% F1-Score, forming an almost flat line near 100% in Fig. 4. A salient observation is Fed-ANIDS: despite its very high accuracy of 99.64%, its Precision, Recall, and F1-Score decline to 49.82%, 50.00%, and 49.91%, respectively, indicating that it is heavily biased toward the majority class. In contrast, Falcon maintains both high accuracy and well-balanced error rates, outperforming all FL-based methods and even the centralized TMG-GAN.

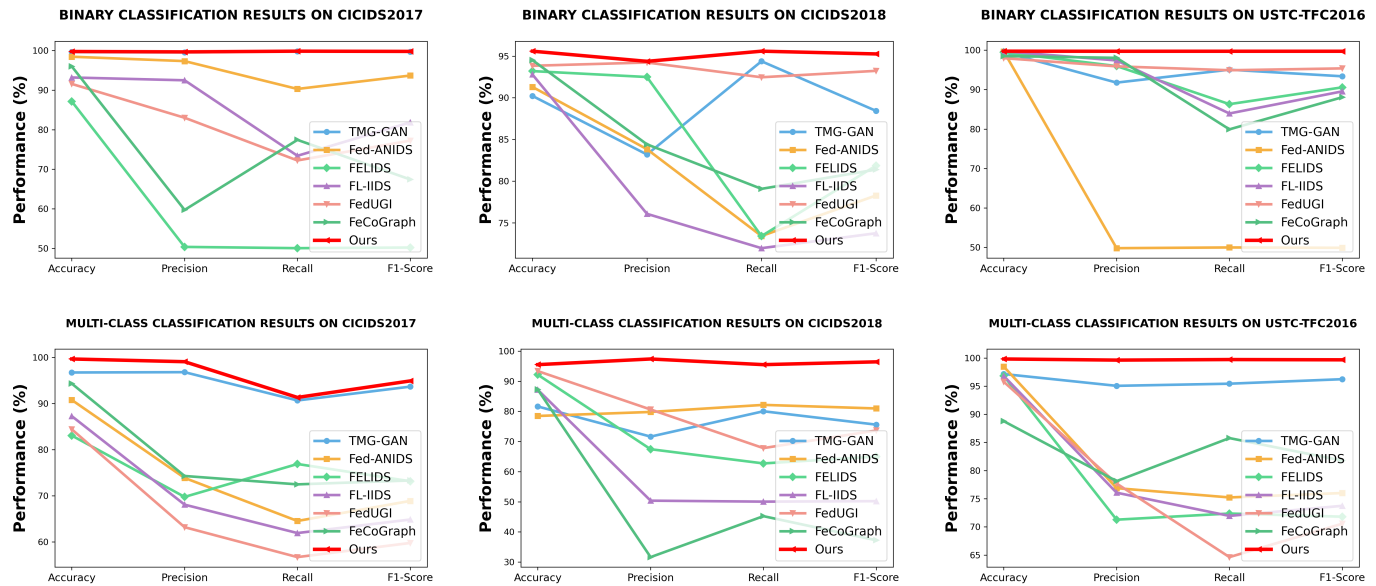


Fig. 4. Overall binary (top) and multi-class (bottom) classification performance (Accuracy, Precision, Recall, F1-Score) of different methods on *CICIDS2017*, *CICIDS2018*, and *USTC-TFC2016*.

On *CICIDS2017*, TMG-GAN serves as a strong centralized upper bound with 99.63% Accuracy and 99.72% F1-Score. Falcon still slightly improves upon this baseline, achieving 99.77% Accuracy, 99.67% Precision, 99.85% Recall and 99.79% F1-Score. Other FL baselines suffer from either low precision or low recall: for example, FELIDS degrades to about 50% on all three metrics (Precision, Recall and F1-Score), and FeCoGraph reaches only 67.42% F1-Score although its accuracy is 96.00%. This behavior is reflected by the steep declines from the Accuracy point to the Recall and F1-Score points in Fig. 4, whereas Falcon remains close to the upper boundary on every metric.

On *CICIDS2018*, Falcon attains 95.59% Accuracy, 94.37% Precision, 95.59% Recall and 95.26% F1-Score, preserving a good balance between Precision and Recall. FedUGI is the strongest competitor in this case with 93.84% Accuracy and 93.23% F1-Score, while TMG-GAN yields 88.43% F1-Score. Several baselines again exhibit strong metric imbalance: Fed-ANIDS achieves 91.30% Accuracy but only 78.27% F1-Score, and FeCoGraph attains 94.52% Accuracy yet only 81.38% F1-Score. Overall, across the three datasets, Falcon not only eliminates the performance gap between privacy-preserving FL and centralized training, but also yields strictly higher F1-Scores than TMG-GAN on all datasets.

Multi-class Classification Experiments: We further evaluate all methods in the multi-class setting. The results in Fig. 4 lead to similar conclusions: Falcon again provides the most stable and superior performance on all datasets, and its curves exhibit the smallest decline from Accuracy to F1-Score. This effect becomes more pronounced in the multi-class setting, where class imbalance is more severe. The combination of balanced sampling and the loss reweighting mechanism in Eq. (4) prevents the classifier from being dominated by majority classes, thereby improving the discriminative capability for minority and long-tailed attack categories. On *USTC-*

TFC2016, Falcon achieves **99.82%** Accuracy, **99.62%** Precision, **99.72%** Recall and **99.67%** F1-Score, clearly surpassing the centralized TMG-GAN (97.15% Accuracy and 96.23% F1-Score). In contrast, Fed-ANIDS, FELIDS, FL-IIDS and FedUGI obtain F1-Scores of 76.01%, 71.81%, 73.74% and 70.56%, respectively, and FeCoGraph reaches 81.77% F1-Score. The pronounced declines of their curves from Accuracy to Recall/F1-Score in Fig. 4 show that these FL-based baselines struggle with the imbalanced multi-class malware traffic, while Falcon maintains almost perfectly horizontal performance curves close to 100%.

On *CICIDS2017*, TMG-GAN again constitutes a strong non-federated baseline, with 96.73% Accuracy and 93.66% F1-Score. Falcon attains 99.66% Accuracy and 99.07% Precision, with 91.32% Recall and an F1-Score of **94.90%**, thereby outperforming TMG-GAN on all four metrics. Among the FL baselines, FeCoGraph is the best, but its F1-Score is only 73.31%, and the remaining FL methods stay below 69%. Thus, even in the more challenging multi-class scenario, Falcon substantially narrows and in fact reverses the performance gap between federated and centralized paradigms.

On *CICIDS2018*, Falcon maintains robust performance with **95.50%** Accuracy, **97.38%** Precision, **95.50%** Recall and **96.43%** F1-Score. By comparison, TMG-GAN attains 81.62% Accuracy and 75.59% F1-Score, and the strongest FL baseline, Fed-ANIDS, reaches 78.47% Accuracy and 80.96% F1-Score. Several FL methods show severe degradation on minority classes: for instance, FeCoGraph still reports 87.22% Accuracy, but its Precision, Recall and F1-Score decline to 31.63%, 45.22% and 37.22%, respectively, which is visualized as a sharp decline of its curve in Fig. 4. Falcon, on the other hand, preserves both high Recall and high Precision across all classes, resulting in the highest F1-Score among all methods.

In summary, TMG-GAN is included as a strong centralized NIDS baseline that assumes full access to globally aggregated

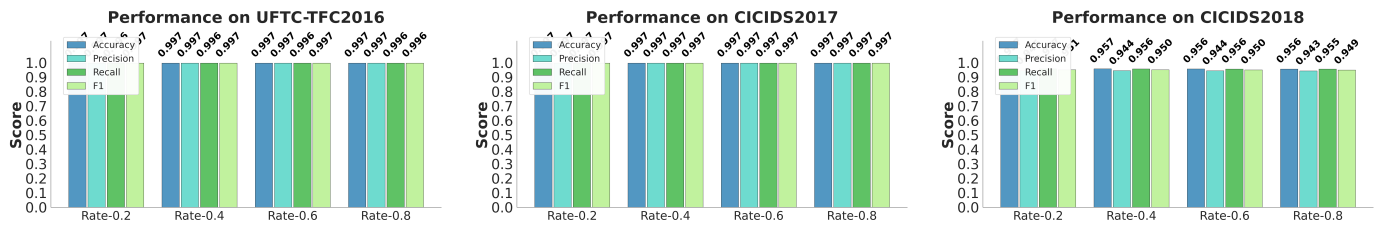


Fig. 5. Robustness evaluation under client dropout scenarios across three datasets (*USTC-TFC2016*, *CICIDS2017*, *CICIDS2018*) under varying dropout rates.

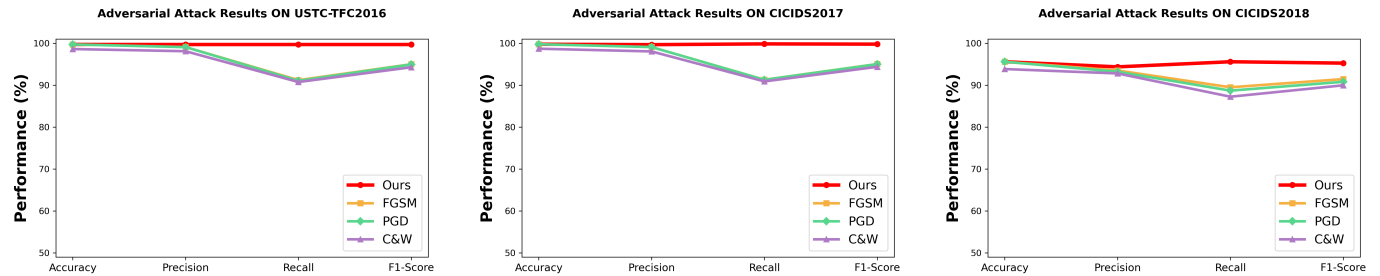


Fig. 6. Robustness evaluation under adversarial attacks across three datasets (*USTC-TFC2016*, *CICIDS2017*, *CICIDS2018*) under FGSM attacks with varying perturbation magnitudes.

traffic at a single site. This baseline allows us to directly examine whether a privacy-preserving federated design must compromise detection performance compared with a centralized NIDS. Across all three datasets and in both binary and multi-class settings, Falcon consistently achieves higher F1-Scores and comparable or better accuracy than this centralized baseline, while keeping raw traffic data on each client and exchanging only model updates. These results show that the proposed federated framework not only outperforms existing FL-based NIDS, but also surpasses a state-of-the-art centralized NIDS, indicating that competitive or superior intrusion-detection performance can be attained without centralized training on raw network flows.

Client Dropout Experiments: In real-world federated learning deployments, client availability is often unstable due to network interruptions, device failures, or resource constraints. To evaluate the robustness of Falcon under such conditions, we conduct systematic experiments with client dropout rates ranging from 0.2 to 0.8, as illustrated in Fig. 5.

On *USTC-TFC2016*, Falcon demonstrates exceptional stability across all dropout rates, with performance metrics remaining consistently above **99.6%**. On *CICIDS2017*, similar robustness is observed with only **0.03%** degradation from rate 0.2 to 0.8. On *CICIDS2018*, despite more complex traffic patterns, performance degradation remains within **0.2%** across all metrics. These results validate that Falcon maintains robust intrusion detection performance under realistic federated scenarios where client participation is unstable.

Adversarial Attack Experiments: To assess the security robustness of Falcon against adversarial perturbations, we conduct systematic evaluations using the Fast Gradient Sign Method (FGSM) with perturbation magnitudes $\epsilon \in \{0.01, 0.05, 0.1, 0.2\}$, as illustrated in Fig. 6.

On *USTC-TFC2016*, Falcon exhibits strong resilience with performance metrics remaining above **99.64%** even under the

strongest attacks. On *CICIDS2017*, performance degradation is less than **0.05%** across all perturbation magnitudes. On *CICIDS2018*, despite more complex traffic patterns, the maximum degradation remains within **0.2%**. These results validate that Falcon's architecture provides inherent defense against gradient-based attacks, maintaining reliable intrusion detection even when facing adversarially crafted malicious traffic.

Focal Loss Parameter Sensitivity: To validate the robustness of hyperparameter configuration, we conduct sensitivity analysis on Focal Loss parameters: gamma (γ), alpha (α), and tau (τ), as illustrated in Fig. 7.

Across all datasets, the model demonstrates stable performance under varying parameter settings. Performance remains consistent for gamma values from 0 to 5, alpha values between 0.1 and 0.75, and tau values from 0.01 to 0.2. These results confirm that Falcon maintains reliable detection performance across a wide range of hyperparameter configurations.

Non-IID Heterogeneity Analysis: To evaluate Falcon's robustness under heterogeneous data distributions, we conduct systematic experiments using Dirichlet-based *non-IID* partitioning with varying concentration parameter $\alpha \in \{1, 2, 3, 4, 5\}$, as illustrated in Fig. 8. Lower α values indicate stronger statistical heterogeneity across clients.

On *USTC-TFC2016*, even under severe heterogeneity ($\alpha = 1$), Falcon maintains over **97.4%** performance across all metrics, with only **2.3%** degradation compared to the IID baseline. As α increases to 5, performance converges to within **0.14%** of the IID setting. On *CICIDS2017*, similar trends are observed with **2.2%** maximum degradation at $\alpha = 1$ and near-IID performance at $\alpha = 5$. On *CICIDS2018*, despite more complex traffic patterns, Falcon achieves over **90.5%** F1-Score even under $\alpha = 1$, demonstrating strong resilience to data heterogeneity. These results validate that Falcon's federated rebalancing strategy and robust feature extraction effectively mitigate the adverse impact of *non-IID* data distributions in

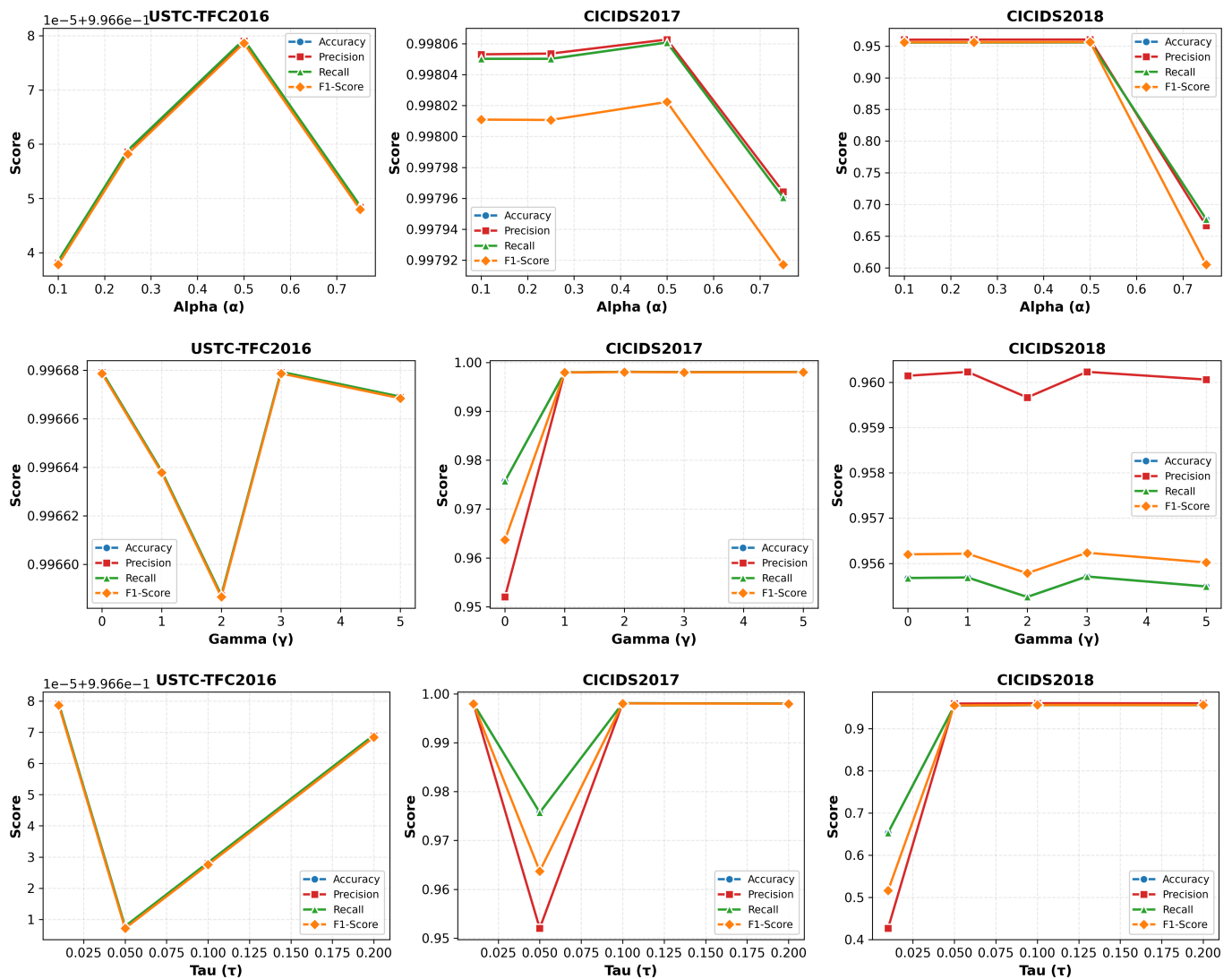


Fig. 7. Parameter sensitivity analysis for Focal Loss hyperparameters across three datasets.

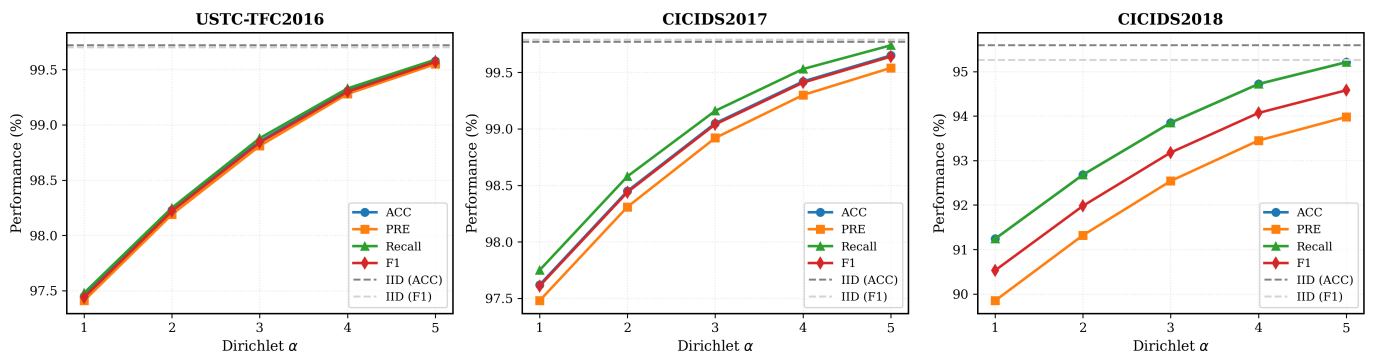


Fig. 8. Impact of data heterogeneity on model performance under varying Dirichlet α values. Lower α indicates stronger *non-IID* conditions. The horizontal dashed lines represent IID baseline performance. As α increases, performance converges toward the IID baseline, demonstrating Falcon's robustness under heterogeneous federated settings.

real-world federated intrusion detection scenarios. From a theoretical perspective, this robustness can be explained by the joint optimization design of Falcon. The balanced sampler ensures a more uniform class distribution within each mini-

batch as described in Eqs. (1)–(2), while the Focal Loss in Eq. (4) further increases the contribution of hard samples during training. As a result, local model updates are less biased toward client-specific majority classes, leading to improved

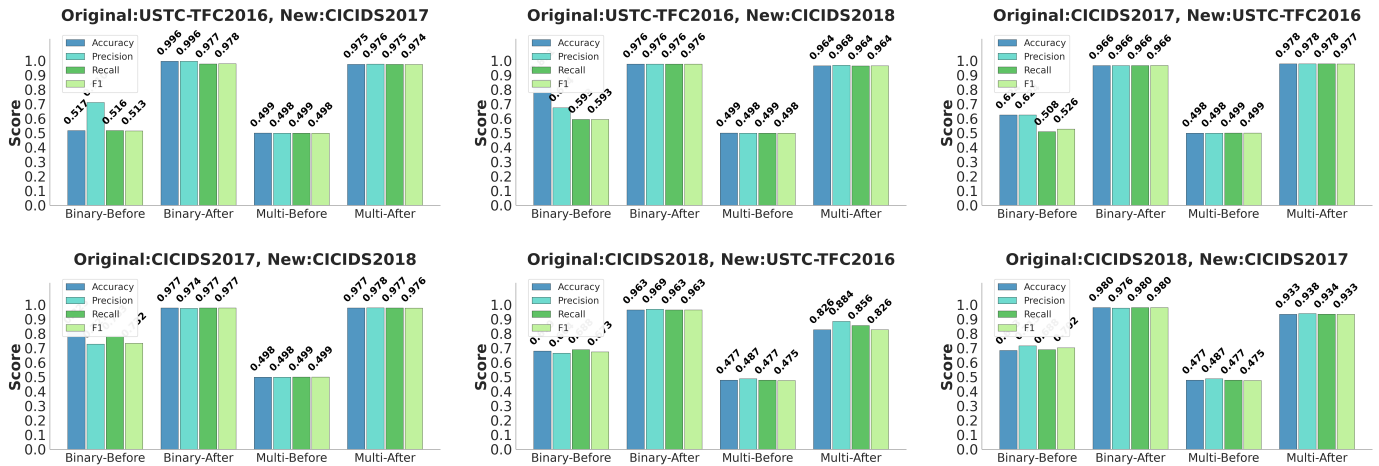


Fig. 9. Experimental results of incremental learning under cross-dataset drift scenarios. The figure illustrates the performance of the proposed framework before and after incremental learning across different dataset combinations. Each subfigure represents a specific transfer pair, where the "Original" dataset serves as the base model training set and the "New" dataset represents novel traffic introduced after drift. The results include binary and multi-class classification tasks, evaluated in terms of Accuracy, Precision, Recall, and F1-score. It can be observed that after the incremental update, the model effectively adapts to new traffic distributions while maintaining stable detection accuracy on previously learned datasets.

global aggregation under heterogeneous data distributions.

Incremental Learning Experiments: As illustrated in Fig. 9, the model exhibits clear and consistent improvements after adapting to new datasets across both binary and multi-class classification tasks. When using *USTC-TFC2016* as the original dataset and *CICIDS2017* as the new dataset, the binary classification metrics—Accuracy, Precision, Recall, and F1-Score—are initially low before transfer, but all approach or exceed **0.97** after incremental training, demonstrating strong adaptability and generalization. Similarly, in the multi-class task, performance rises dramatically from near-random levels (around **0.49**) to approximately **0.97**, indicating that the proposed incremental learning strategy effectively mitigates catastrophic forgetting. When extended to other dataset combinations, such as those involving *CICIDS2018*, the model continues to maintain high Accuracy and Recall, with F1-Score consistently within the **0.96–0.98** range, demonstrating its stability under complex distribution shifts. The effectiveness of the incremental adaptation can be explained by the KL-based drift detection mechanism. As defined in Eq. (19), the KL divergence quantifies the discrepancy between the reference distribution and the current data distribution. Once the divergence exceeds a predefined threshold, the model is updated accordingly, enabling Falcon to adapt to newly emerging attack patterns in a timely manner. Furthermore, the strong knowledge retention can be explained by the multi-teacher distillation framework in Eqs. (22)–(24). By constraining the student model to align with both historical and new-class teacher outputs, and combining this with memory replay, Falcon effectively balances stability and plasticity, thereby mitigating catastrophic forgetting during incremental updates.

Furthermore, knowledge retention is evaluated during cross-dataset incremental learning, as illustrated in Fig. 10. For binary classification, retention typically remains around **99%**, and in some cases even exceeds **100%**, suggesting that the model not only preserves but also reinforces previously learned

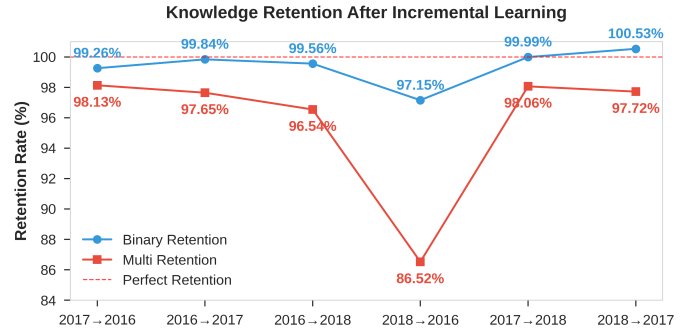


Fig. 10. Knowledge retention after incremental learning. Retention rates of binary and multi-class models on previously learned data under different incremental learning orders (2017→2016, 2016→2017, etc.), compared against an ideal perfect-retention baseline.

knowledge. For multi-class classification, although minor fluctuations are observed, retention remains stable within **96%–98%**, with only slight declines in isolated cases. Overall, these results confirm that the proposed method achieves robust and scalable cross-domain incremental learning, effectively absorbing new knowledge while preserving prior information.

Ablation Study: This section evaluates the contribution of each module via ablation experiments: we use a vanilla LSTM as the baseline, our proposed model as the improved architecture, and then progressively introduce class-imbalance handling (denoted as *Balance*). As shown in Table II, the standalone LSTM is unstable under imbalanced settings—for example, on *USTC-TFC2016* (binary) it yields only **11.35%** Accuracy with an F1-Score of **3.90%**; the multi-class results on *CICIDS2018* are likewise low. Adding *Balance* on top of the baseline alleviates missed detections and improves Recall substantially, but often leads to a skewed “high-precision, low-recall” trade-off, so the overall F1-Score remains limited.

Replacing LSTM with our proposed model, even *without Balance*, produces markedly more stable decision

TABLE II
ABLATION RESULTS ON BINARY AND MULTI-CLASS CLASSIFICATION

Datasets	Combination	Balance	Model		Binary Metrics				Multi Metrics			
			Ours	LSTM	ACC	PRE	REC	F1	ACC	PRE	REC	F1
USTC-TFC2016	1		✓		62.96	49.70	98.26	66.03	77.36	44.00	27.00	27.00
	2			✓	11.35	3.10	13.30	3.90	36.67	18.33	50.00	26.84
	3	✓		✓	63.65	99.99	0.88	1.74	63.43	8.90	9.24	7.34
	4	✓	✓		99.72	99.70	99.70	99.70	99.82	99.62	99.72	99.67
CICIDS2017	1		✓		97.41	0	0	0	97.57	19.51	20.00	19.75
	2			✓	92.02	28.55	58.38	32.57	97.57	0	0	0
	3	✓		✓	95.98	38.73	80.23	24.88	89.20	13.20	61.90	21.80
	4	✓	✓		99.77	99.67	99.85	99.79	99.66	99.07	91.32	94.90
CICIDS2018	1		✓		35.01	34.76	99.68	51.55	65.31	20.00	20.00	20.00
	2			✓	34.69	34.69	99.99	51.51	25.33	20.44	20.44	19.42
	3	✓		✓	61.50	73.42	70.43	61.21	23.46	21.42	40.00	7.93
	4	✓	✓		95.59	94.37	95.59	95.26	95.50	97.38	95.50	96.43

boundaries and tangible gains in F1-Score for both binary and multi-class tasks. When our model operates *with Balance*, it achieves the global optimum. For example, on *USTC-TFC2016* (binary), **Accuracy/Precision/Recall/F1-Score** reach **99.72/99.70/99.70/99.70**; on *CICIDS2017* (multi-class), Accuracy is **99.66%** with an F1-Score of **94.90%**; and on *CICIDS2018* (multi-class), the results further improve to **95.50%** Accuracy and **96.43%** F1-Score. These findings indicate that the vanilla LSTM provides only basic temporal modeling, *Balance* mainly recovers Recall, while our model supplies stable and interpretable discriminative features. No single component alone can simultaneously optimize Precision and Recall. Only the **complete configuration**—our model *plus Balance*—consistently delivers the highest and most balanced metrics across datasets and across both binary and multi-class tasks, demonstrating that our improvements are both necessary and complementary. These observations are consistent with the mathematical design of Falcon. The balanced sampling mechanism modifies the effective training distribution, while the Focal Loss in Eq. (4) adjusts the gradient contribution across classes. Meanwhile, the CNN-BiLSTM-Attention architecture enhances feature representation as defined in Eqs. (8)–(18). Only the combination of these components leads to consistent improvements across all evaluation metrics.

V. CONCLUSION

This paper proposed **Falcon**, an imbalance-aware federated intrusion detection framework for heterogeneous consumer-centric IoT. By integrating data balancing, multi-level feature extraction, and robust federated training, Falcon effectively addresses privacy constraints, class imbalance, and *non-IID* client heterogeneity.

Experiments on three benchmark datasets demonstrate that Falcon consistently improves F1-Score, Recall, and overall detection stability over strong baselines, while remaining robust under *non-IID* data distributions and client dropout. The

ablation study further verifies that each component contributes substantially to the final performance.

Future Work: Future work will explore three promising directions to further enhance Falcon’s robustness and applicability. First, we plan to integrate **lightweight noise detection** mechanisms by combining uncertainty quantification (e.g., Monte Carlo dropout or ensemble-based confidence estimation) with client-side data quality scoring, enabling the server to adaptively down-weight noisy client updates during aggregation without violating privacy constraints. Second, we will investigate **hard-example mining** strategies that leverage attention-weighted sample selection and curriculum learning to prioritize challenging minority-class instances during local training, thereby improving the model’s discriminative capability on rare and evolving attack patterns. Third, we aim to develop **communication-efficient incremental adaptation** by exploring gradient compression techniques (e.g., top-k sparsification and quantization) combined with selective model broadcasting, reducing the communication overhead of frequent incremental updates in resource-constrained CIoT environments. These directions are motivated by recent surveys highlighting the necessity of lightweight data quality control and efficient communication for sustaining robustness under heterogeneous and evolving environments [41].

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